

B.Tech Project Report

Video-to-Video Translation

GitHub Repository:

https://github.com/Roshan-o/V_v_website

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1 Introduction

1.1 What is Video-to-Video Translation

Video-to-Video Translation (V2V Translation) is an Artificial Intelligence based technology that converts the spoken language in a video into another language while preserving the speaker's facial expressions, emotions, voice characteristics, and lip movements. Unlike traditional subtitle-based systems, V2V Translation generates a translated video in which the speaker appears to naturally speak the target language, creating a more realistic and engaging viewing experience. This process combines multiple technologies such as Automatic Speech Recognition (ASR), Machine Translation (MT), Text-to-Speech (TTS), voice conversion, lip synchronization, and facial video generation to produce a seamless translated output video.

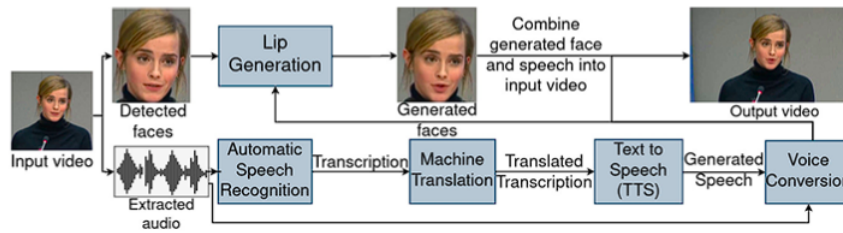


Figure 1: Overview

The system works through multiple stages. First, audio is extracted from the input video and converted into text using Automatic Speech Recognition (ASR). The transcribed text is then translated into the target language using Machine Translation (MT). After translation, Text-to-Speech (TTS) systems generate speech audio in the target language while voice conversion techniques help preserve the speaker's original tone and speaking style. Finally, lip synchronization and facial animation modules align the generated speech with the speaker's mouth movements to produce the final translated video.

Video-to-Video Translation helps overcome language barriers and improves accessibility to multilingual content. It is especially useful in healthcare, where doctor-patient communication may involve different languages, and in education, where lectures and learning materials can be translated into regional languages for better understanding. The technology is also useful in entertainment, business communication, online meetings, and social media content creation.

Overall, Video-to-Video Translation combines speech recognition, translation, speech synthesis, and lip synchronization into a unified system that enables realistic multilingual communication and improves global accessibility to digital content.

1.2 Study of ASR, MT, and TTS Modules

1.2.1 Automatic Speech Recognition (ASR)

Automatic Speech Recognition (ASR) is the first stage of the Video-to-Video Translation pipeline. The main function of ASR is to convert spoken audio from the input video into text. The system processes speech signals, extracts acoustic features, and generates transcription along with timestamp information. The generated text is later used for translation, subtitle generation, and synchronization. ASR accuracy is important because transcription errors directly affect the quality of translation and lip synchronization.

Common ASR Models

- **Whisper** – Multilingual speech recognition model developed by OpenAI. Supports noisy audio and multiple languages with high accuracy.

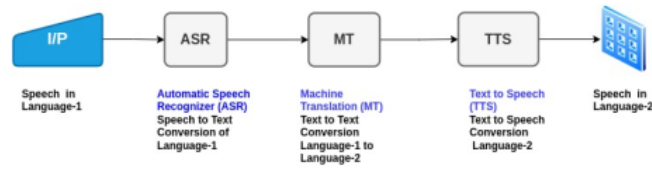


Figure 1.2: Cascaded Speech Translation (I/P stands for Input)

Figure 2: PIPELINE

- **Faster-Whisper** – Optimized version of Whisper with faster inference speed and lower computational cost.
- **wav2vec 2.0** – Self-supervised speech recognition model developed by Meta AI for robust speech understanding.
- **Google Speech-to-Text** – Cloud-based ASR service with real-time multilingual speech recognition.

Key Features

- Speech-to-text conversion
- Timestamp generation
- Noise handling
- Multilingual support
- Speaker segmentation support

1.2.2 Machine Translation (MT)

Machine Translation (MT) converts transcribed text from the source language into the target language while preserving meaning, grammar, and context. Modern MT systems mainly use Neural Machine Translation (NMT) based on transformer architectures. Translation quality is important because incorrect translation affects speech generation and overall communication quality.

Common MT Models

- **IndicTrans** – Translation model designed specifically for Indian languages.
- **mBART** – Multilingual sequence-to-sequence transformer model for translation tasks.
- **MarianMT** – Open-source neural machine translation framework with multilingual support.
- **Google Translate API** – Large-scale cloud translation service supporting many languages.

Key Features

- Multilingual translation
- Context-aware translation
- Transformer-based architecture
- Domain adaptation support
- Grammar and semantic preservation

1.2.3 Text-to-Speech (TTS)

Text-to-Speech (TTS) converts translated text into natural speech audio in the target language. The system predicts pronunciation, pitch, rhythm, and speaking style to generate realistic speech. In Video-to-Video Translation, TTS helps create natural translated audio that can later be synchronized with facial movements and lip synchronization modules.

Common TTS Models

- **Tacotron 2** – Deep learning based speech synthesis model for natural speech generation.
- **FastSpeech 2** – Faster and efficient non-autoregressive TTS model with high-quality output.
- **VITS** – End-to-end neural speech synthesis model capable of generating highly natural speech.
- **gTTS (Google Text-to-Speech)** – Cloud-based speech synthesis service supporting multiple languages.

Key Features

- Natural speech generation
- Emotion and pitch control
- Multilingual voice synthesis
- Voice cloning support
- Real-time speech generation

2 Analysis of Previous Research Work on Video-to-Video Translation

As part of our project, we studied the previous research work and thesis developed by Vandhan to gain a detailed understanding of Video-to-Video Translation systems, their architecture, workflow, and major technical challenges. This analysis helped us understand how modules such as Automatic Speech Recognition (ASR), Machine Translation (MT), Text-to-Speech (TTS), speaker diarization, and lip synchronization work together in a multilingual video translation pipeline.

Through this study, we identified important challenges including speaker identification, gesture understanding, emotion preservation, lip synchronization mismatch, and error propagation across modules. The research also highlighted the importance of multi-modal processing and Human-in-the-Loop systems for generating realistic and accurate translated videos.

The thesis describes Video-to-Video Translation as a system that converts a video spoken in one language into another language while preserving the speaker’s facial expressions, emotions, lip movements, and speaking style. Unlike subtitle-based systems, the translated output appears as if the speaker is naturally speaking in the target language. The system combines technologies such as ASR, MT, TTS, voice conversion, and lip synchronization to generate multilingual translated videos.

From the analysis, we understood the importance of multilingual accessibility, especially in countries like India where language diversity creates communication barriers in healthcare, education, and online services. The research explains how Video-to-Video Translation can help provide lectures, medical communication, and digital content in regional languages such as Telugu, Hindi, and Tamil.

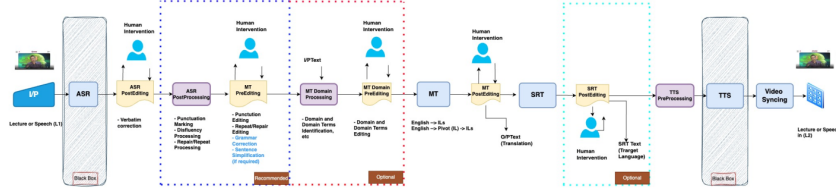


Figure 1.3: Proposed cascaded Speech to Speech Machine Translation

Figure 3: SSMT

The work also analyzed both Direct Speech Translation and Cascaded Speech Translation approaches. The cascaded approach was identified as more practical because it separates the system into independent modules such as Speech-to-Text, Machine Translation, and Text-to-Speech. This modular architecture improves debugging, scalability, and independent optimization of each stage while also supporting Human-in-the-Loop correction mechanisms.

The proposed pipeline in the thesis begins with extracting audio and facial information from the input video. The speech is converted into text using ASR, followed by preprocessing techniques such as punctuation restoration, disfluency removal, and text cleaning. The cleaned transcription is translated into the target language using Machine Translation models. The translated text is then converted into speech using TTS systems while preserving speaker characteristics. Finally, lip synchronization and video alignment modules synchronize translated speech with facial movements to generate realistic translated videos.

The analysis also emphasized the importance of Human-in-the-Loop systems because fully automatic translation systems may produce transcription and translation errors. Human intervention helps correct ASR transcription mistakes, punctuation errors, translation issues, and domain-specific terminology before generating the final translated output. This helps reduce cascading errors across the pipeline.

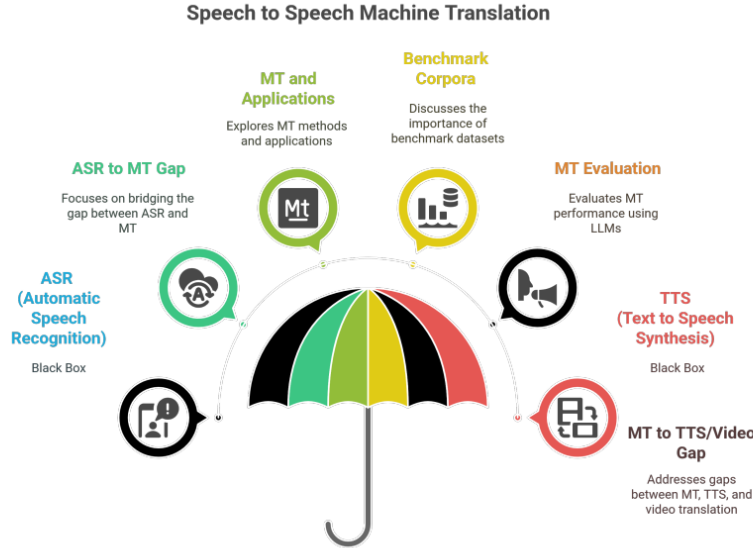


Figure 4: Highlighting the core challenges and components of the Speech-to-Speech Machine Translation (SSMT)

The thesis further explained the complete workflow of Cascaded Speech-to-Speech Machine Translation (SSMT), including input video processing, ASR transcription, Machine Translation, subtitle generation, Text-to-Speech synthesis, lip synchronization, and final translated video generation. System

performance was evaluated using metrics such as Word Error Rate (WER), BLEU Score, and Mean Opinion Score (MOS) to measure transcription accuracy, translation quality, and speech naturalness.

From this analysis, we gained a strong understanding of how modern Video-to-Video Translation systems are designed and the challenges involved in generating realistic multilingual translated videos. These insights guided our further analysis of speaker diarization, gesture understanding, lip synchronization, emotion preservation, and multi-modal fusion techniques for our future Video-to-Video Translation pipeline development.

3 Multi-Modal Fusion and Dataset Analysis

3.1 Understanding Multi-Modal Video Translation

As part of our Video-to-Video Translation project, we analyzed the research paper “*TRAVID: An End-to-End Video Translation Framework*” to understand how multi-modal learning and fusion techniques improve multilingual video translation systems. The paper explains how different modalities such as speech audio, video frames, facial expressions, lip movements, gestures, and text information work together to improve synchronization, translation quality, and speaker consistency.

From this analysis, we understood that Video-to-Video Translation is not only a speech translation problem but also a multi-modal synchronization problem where both audio and visual information must be processed together to generate realistic translated videos.

3.2 Datasets Studied

During our study, we analyzed several datasets commonly used in audio-visual speech processing and lip synchronization systems. Datasets such as LRS2, LRS3, LRW, and ReSyncED contain real-world talking face videos with different speakers, lighting conditions, face poses, and speaking styles. These datasets are widely used for lip reading, audio-visual speech recognition, lip synchronization, and visual speech understanding.

LRS2 and LRS3 mainly focus on large-scale talking face videos and are commonly used for training speech-driven lip synchronization systems. LRW is useful for visual speech recognition and word-level lip analysis, while ReSyncED is mainly used for evaluating real-world lip synchronization quality and audio-video consistency. Through these datasets, we understood the importance of diverse real-world video data for training robust multilingual translation systems.

3.3 Fusion Techniques and Multi-Modal Learning

The paper also explains the importance of multi-modal fusion techniques in Video-to-Video Translation systems. Multi-modal fusion refers to combining different modalities such as audio, video, lip movements, facial expressions, and text features inside a single system to improve synchronization and overall translation quality.

During our analysis, we studied three major fusion approaches: Early Fusion, Late Fusion, and Hybrid Fusion. Early Fusion combines audio, visual, and text features at the initial stage itself so that the model can jointly learn relationships between different modalities. This approach improves synchronization learning but increases computational complexity because large feature representations are processed together.

Late Fusion processes each modality independently and combines their outputs only during the final prediction stage. This approach improves modularity and robustness because errors in one modality do not completely affect the entire system. However, it provides weaker interaction between modalities compared to Early Fusion.

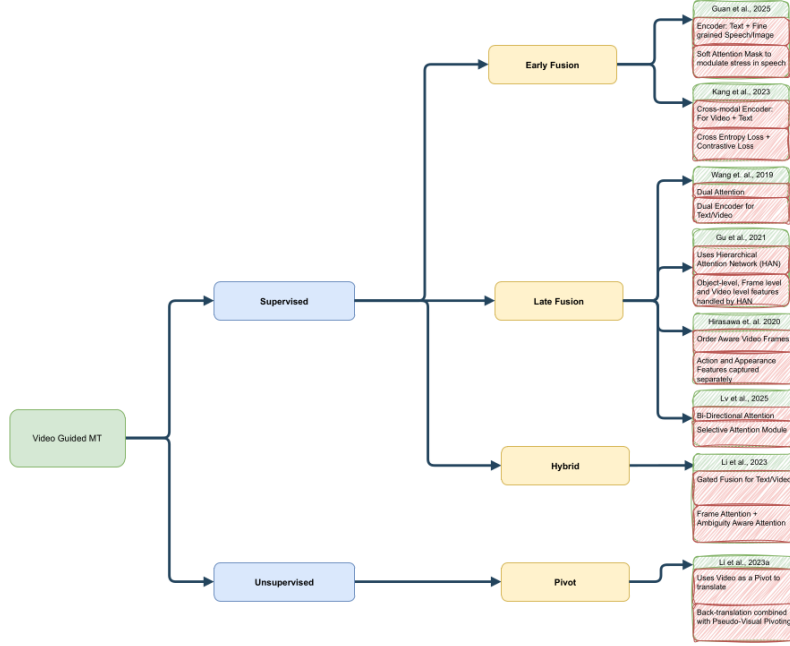


Figure 5: Taxonomy for Video Guided Machine Translation

Hybrid Fusion combines both Early Fusion and Late Fusion approaches to balance synchronization quality and computational efficiency. Important modalities such as speech audio and lip movements are fused early, while translation and speaker identification modules can be processed independently before final integration. From this analysis, we understood that Hybrid Fusion provides a more scalable and practical architecture for real-world Video-to-Video Translation systems.

This study helped us understand how multiple modalities such as speech, facial expressions, gestures, lip movements, and visual cues contribute to realistic multilingual communication. We learned that proper fusion techniques improve audio-video synchronization, speaker consistency, translation alignment, gesture understanding, and lip synchronization quality.

The concepts learned from the TRAVID framework and multi-modal fusion techniques will help improve speaker tracking, synchronization, gesture-aware translation, and overall realism in our future Video-to-Video Translation pipeline.

4 Key Challenges Identified in Translation

During the analysis and development of our Video-to-Video Translation pipeline, we identified several major challenges that directly affect translation quality, synchronization, realism, and overall user experience. These challenges were studied to understand how real-world multilingual video translation systems operate and what improvements are required for future integration.

4.1 Speaker Diarization Challenges

Speaker diarization is the process of identifying “who spoke when” in a conversation. In multi-speaker videos such as interviews, meetings, classrooms, and doctor-patient conversations, separating speakers accurately becomes difficult. Overlapping speech, background noise, and similar voice patterns can confuse the system and lead to incorrect speaker identification.

Incorrect speaker mapping creates problems such as wrong subtitle assignment, incorrect translated voice generation, improper lip synchronization, and loss of conversational flow. To address this challenge,

we analyzed audio-based speaker diarization systems using speaker embeddings, clustering methods, and multi-modal approaches combining voice and visual information.

4.2 Emotion Preservation Challenges

Human communication contains emotional information such as happiness, sadness, anger, excitement, and sarcasm. Traditional Machine Translation systems mainly focus on textual meaning and often fail to preserve emotional context during translation. As a result, emotional intensity may be reduced and translated speech may sound neutral or unnatural.

Similarly, Text-to-Speech systems may generate robotic or emotionless speech even when the original speaker expresses strong emotions. This reduces the realism and naturalness of translated videos. We identified emotion preservation as an important challenge because realistic multilingual communication requires maintaining both semantic meaning and emotional expression.

4.3 Gesture and Facial Expression Challenges

Human communication also depends heavily on non-verbal information such as facial expressions, eye movements, hand gestures, head movements, and body posture. These visual cues provide additional context and improve communication understanding. However, many translation systems mainly focus on speech and ignore gesture-related information.

A major challenge occurs when translated speech timing changes while gestures and facial movements remain aligned with the original speech. This creates inconsistencies between speech and body language. To better understand this problem, we studied multi-modal systems that combine audio features, facial information, and body movement analysis for preserving communication realism during translation.

4.4 Lip Synchronization Challenges

Lip synchronization is one of the most difficult components in Video-to-Video Translation. After translation and speech generation, the speaker’s lip movements must be modified to match the translated speech audio. This becomes difficult because different languages have different sentence structures, speaking speeds, and mouth movement patterns.

Speech timing changes after translation can create synchronization mismatches, especially during fast head movements or side-face views where lip visibility becomes limited. Even small synchronization errors are easily noticeable and reduce video realism. To understand this challenge, we analyzed lip synchronization models such as Wav2Lip and studied audio-visual synchronization techniques for future integration into our pipeline.

4.5 Multi-Speaker Video Challenges

Most existing translation and lip synchronization systems perform well mainly for single-speaker videos. However, real-world videos often contain multiple speakers, frequent speaker switching, simultaneous conversations, and dynamic camera angles. These situations make synchronization and speaker tracking significantly more difficult.

Handling multi-speaker videos requires accurate face tracking, voice association, speaker clustering, and synchronization between translated speech and visual appearance. Through our analysis, we identified multi-speaker handling as a major research challenge in realistic Video-to-Video Translation systems.

5 Research Analysis of Identified Challenges

5.1 Speaker Diarization Analysis

As part of our analysis, we studied speaker diarization techniques to understand how multi-speaker conversations can be handled in Video-to-Video Translation systems. Speaker diarization is the process of identifying and separating different speakers in an audio or video recording. In real-world conversations such as interviews, meetings, classrooms, and doctor-patient interactions, multiple speakers may speak at different times or simultaneously, making speaker separation an important challenge before translation and lip synchronization.

During the analysis, we identified that incorrect speaker mapping can create problems such as wrong transcription, incorrect subtitle assignment, improper translated voice generation, and lip synchronization mismatch. To understand this problem, we studied both text-based and audio-based diarization approaches. Text-based diarization groups speakers using sentence structure and conversational patterns after speech-to-text conversion, but it becomes unreliable when ASR errors or overlapping speech occur.

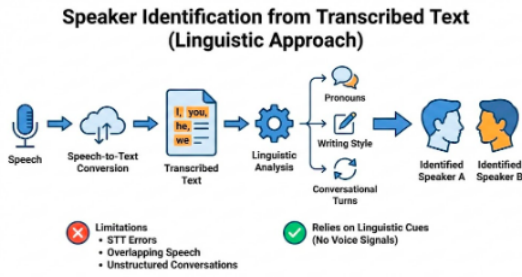


Figure 6: Text-Based Diarization



Figure 7: Audio-Based Diarization

We found audio-based diarization to be more practical because it identifies speakers directly using acoustic voice features and speaker embeddings. We analyzed diarization pipelines involving audio pre-processing, speech segmentation, speaker embedding extraction, and clustering techniques. Models and libraries such as Whisper, Faster-Whisper, SpeechBrain, pyannote.audio, ECAPA-TDNN, and Agglomerative Clustering were studied to understand how speech segments are grouped and assigned to different speakers.

Through this analysis, we understood the importance of speaker consistency, timestamp mapping, and multi-speaker synchronization in multilingual video translation systems. We also identified challenges such as overlapping speech, background noise, similar voice characteristics, and low-quality audio that affect diarization performance. This study helped us understand how speaker tracking and voice association can improve realistic multi-speaker Video-to-Video Translation.

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5.2 Gesture and Video Feature Analysis

We also analyzed research related to gesture understanding and video feature extraction to study how non-verbal communication affects Video-to-Video Translation systems. Human communication depends not only on speech but also on facial expressions, body posture, hand gestures, head movements, and visual interaction cues. Ignoring these visual features may reduce realism and natural communication flow in translated videos.

As part of this study, we analyzed the paper *“Face, Body, Voice: Video Person-Clustering with Multiple Modalities”*, which explains how multiple modalities such as face features, body appearance,

gestures, and voice information can be combined for person understanding in videos. We understood that relying only on face detection is insufficient because real-world videos often contain side-face views, hidden faces, dynamic camera angles, and continuous speaker movement.



Figure 1: Video Person-Clustering – an essential step towards story understanding. Imagine trying to understand the story in the scenes above, given only the *non-greyled-out* parts. Face-level understanding (left) omits important information, such as characters with their backs turned. This work addresses the new task of video person-clustering, which develops *person-level* understanding (right) in a scene by clustering all people, regardless of if their faces are showing or not. This is in contrast to the more limited, established, task of face-clustering. Person-level understanding is essential for downstream applications of grouping-by-identity such as story understanding, and cannot be achieved by face-clustering alone.

Figure 8: Video details Capturing

The paper proposes a Multi-Modal High-Precision Clustering system that combines face embeddings, voice embeddings, body appearance information, and posture features to maintain speaker identity across video frames. We studied feature extraction techniques using models such as SENet-50, ResNet50, Thin-ResNet-34, and Person Re-Identification models for extracting facial, body, and voice embeddings.

From this analysis, we understood how gestures and body movement information help improve speaker tracking, synchronization, conversational understanding, and video realism. The study also showed how multi-modal learning improves person identification even when one modality fails. These concepts are important for maintaining realistic multilingual communication in future Video-to-Video Translation systems.

5.3 Emotion Preservation Analysis

As part of our research, we studied emotion preservation techniques in multilingual speech translation systems. Human speech contains emotional information such as happiness, sadness, anger, excitement, and calmness, which strongly influence communication meaning. Traditional translation systems mainly focus on textual meaning and often fail to preserve emotional expression during translation.

We analyzed the research paper “*Real-Time Speech-To-Speech Translation with Emotion Preservation*” to understand how emotional context can be preserved in translated speech. The proposed system

combines Automatic Speech Recognition, Speech Emotion Recognition, Neural Machine Translation, and Emotion-Aware Voice Conversion to generate translated speech while maintaining the speaker’s emotional expression.

Through this analysis, we studied how speech features such as pitch, rhythm, prosody, energy variation, and speaking speed are used to identify emotional states. These emotional features are later used during speech synthesis to generate more expressive and natural translated speech. We understood that emotion preservation is important because emotionless or robotic speech reduces realism and communication quality in multilingual conversations.

This study helped us understand the importance of emotion-aware translation systems and how emotional context can improve realism, speaker authenticity, and overall user experience in future Video-to-Video Translation pipelines.

5.4 Lip Synchronization Analysis

Lip synchronization is one of the most important and difficult components in Video-to-Video Translation systems. After translation and speech generation, the speaker’s lip movements must be synchronized with the translated speech audio to generate realistic multilingual videos. Even small mismatches between speech and lip movements can make the translated video appear unnatural.

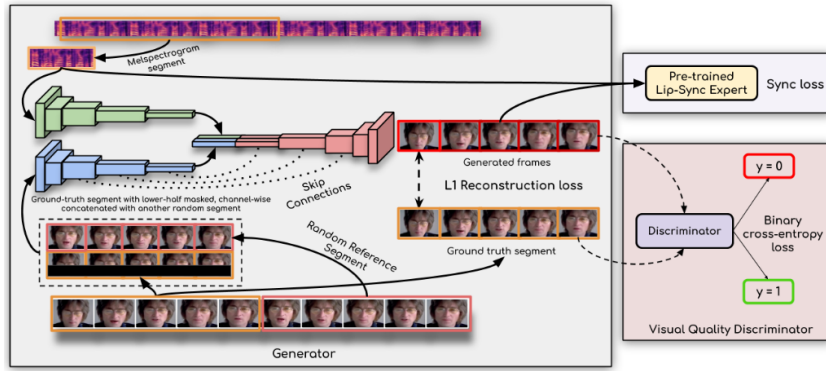


Figure 9: Generates accurate lip-sync by learning from an “already well-trained lip-sync expert”.

To understand this challenge, we analyzed the research paper “*A Lip Sync Expert Is All You Need for Speech to Lip Generation in the Wild*”, which introduces the Wav2Lip model for speech-driven lip synchronization. The paper explains how speech audio can be used to generate synchronized lip movements while preserving facial identity and visual realism.

We studied the architecture of Wav2Lip, which consists of an Identity Encoder, Speech Encoder, Face Decoder, and SyncNet Lip-Sync Expert. The Identity Encoder preserves facial structure and pose consistency, while the Speech Encoder extracts phoneme and timing information from speech audio. The Face Decoder combines these features to generate synchronized lip movements.

The paper also introduces SyncNet, a pretrained lip-sync discriminator that evaluates synchronization between speech audio and lip movements using audio-video embeddings. We analyzed datasets such as LRS2, LRS3, LRW, and ReSyncED, which are commonly used for training and evaluating lip synchronization systems.

From this study, we understood that lip synchronization is an audio-visual synchronization problem that requires accurate timing alignment, phoneme matching, facial consistency, and temporal continuity. These concepts helped us understand how realistic multilingual video generation can be improved using speech-driven lip synchronization techniques.

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6 Model Comparison and Analysis

This section provides a comparative analysis of various Machine Translation (MT), Text-to-Speech (TTS), and Automatic Speech Recognition (ASR) models studied during our project. The comparison mainly focuses on Telugu language support, multilingual capability, offline usability, contextual understanding, and suitability for Video-to-Video Translation systems.

6.1 Machine Translation (MT) Models

Model	Accuracy	Telugu Support	Context Understanding	Offline	Cost	Best Use
Google Translate	High	Good	Moderate	No	Free/Paid API	Fast, general use
DeepL Translator	Very High	Weak	Excellent	No	Paid	European languages
NLLB (Meta)	Very High	Very Good	Good	Yes	Free	Best open-source multilingual
IndicTrans2	High	Excellent	Good	Yes	Free	Selected: Best for Indian languages
Sarvam AI MT	High	Excellent	Good	Partial	Free/Paid	Real-world Indic applications
mBART	Medium-High	Moderate	Moderate	Yes	Free	Research use

Table 1: Comparison of Machine Translation Models

6.2 Text-to-Speech (TTS) Models

Model	Voice Quality	Telugu Support	Voice Cloning	Offline	Cost	Best Use
XTTS	Very High	Weak	Yes	Yes	Free (OS)	Multilingual voice cloning research
Bark	High	Moderate	Limited	Yes	Free (OS)	Expressive speech generation
Sooktam2	High	Excellent	Yes	Yes	Free (OS)	Selected: Telugu voice cloning (computationally heavy)
IndicF5	Very High	Excellent	Yes	Yes	Free (OS)	Selected: Indic multilingual voice cloning (computationally heavy)
Svara TTS	High	Excellent	No	Yes	Free/Limited	Indic offline TTS generation
Sarvam AI TTS	High	Excellent	Limited	No	Paid	Production Indic applications
Google TTS	Very High	Good	No	No	Paid API	Scalable production systems
Tacotron 2	Medium	Weak	No	Yes	Free (OS)	Research and learning

Table 2: Comparison of Text-to-Speech Models

6.3 Automatic Speech Recognition (ASR) Models

Model	Accuracy	Telugu Support	Real-Time	Offline	Cost	Best Use
OpenAI Whisper	High	Good	Limited	Yes	Free (OS)	Selected: Best overall choice
Google STT	Very High	Moderate	Yes	No	Paid API	Production + real-time
wav2vec 2.0	Very High	Needs training	Possible	Yes	Free	Custom ASR research
Vosk	Medium	Weak	Yes	Yes	Free	Lightweight offline applications
DeepSpeech	Low-Medium	Weak	Moderate	Yes	Free	Older ASR systems

Table 3: Comparison of Automatic Speech Recognition Models

7 Implementation Details: Video to Video Translation Website

7.1 Backend Implementation

The backend of the Video-to-Video Translation system was implemented using Python and FastAPI. FastAPI was selected because of its high performance, asynchronous support, and ability to efficiently handle long-running video processing tasks. The system uses asynchronous background processing to execute the translation pipeline without blocking the frontend interface, allowing users to continuously monitor the processing stages in real time.

7.1.1 Core Translation Pipeline

The backend pipeline consists of Automatic Speech Recognition (ASR), Machine Translation (MT), and Text-to-Speech (TTS) modules. Different pretrained models were analyzed and integrated to improve multilingual translation quality and synchronization.

Module	Model Used	Reason for Selection	Key Features
ASR (Automatic Speech Recognition)	OpenAI Whisper (Base)	High transcription accuracy, multilingual support, and robustness to noisy audio conditions.	Timestamp generation, multilingual speech recognition, and reliable transcription across accents.
MT (Machine Translation)	IndicTrans2	Optimized for Indian language translation and better handling of regional language structures.	Supports Indian languages, context-aware translation, and script-specific language handling.
TTS (Text-to-Speech)	XTTS v2 and SvaraTTS	Provides natural speech generation and supports voice style preservation for Indian languages.	Voice cloning, natural speech synthesis, speaker identity preservation, and multilingual support.

Table 4: Core Models Used in the Translation Pipeline

The implemented pipeline follows a segment-based processing approach instead of processing the

complete video at once. Audio is first extracted from the input video and divided into smaller timestamped segments using Whisper. This helps maintain synchronization between translated speech and the original video timeline.

The translated speech segments are merged using `pydub`, where generated audio is placed at the exact timestamps corresponding to the original speech. FFmpeg is used throughout the pipeline for audio extraction, format conversion, and merging the final translated audio back into the original video while preserving video quality.

To manage long-running processing tasks, a UUID-based job management system was implemented. This system tracks the status of every stage such as extraction, transcription, translation, and dubbing while storing intermediate outputs for inspection and debugging.

7.2 Frontend Implementation

The frontend of the system was implemented using React.js and TypeScript. The interface was designed to provide a simple and interactive workflow for uploading videos, tracking pipeline progress, and viewing intermediate outputs generated during processing.

The frontend continuously communicates with the backend using asynchronous polling mechanisms to retrieve the latest processing status. Users can inspect outputs generated at each stage of the pipeline, including extracted audio, transcription results, translated text, and final dubbed video output.

A modular preview system was implemented to allow users to visualize each stage independently. The system also includes a side-by-side comparison view where the original video and translated video can be viewed simultaneously for evaluating synchronization and translation quality.

7.2.1 Website Features

The developed website supports uploading video formats such as MP4 and MOV for multilingual processing. The interface displays live pipeline tracking information showing the status of each processing stage including transcription, translation, and dubbing.

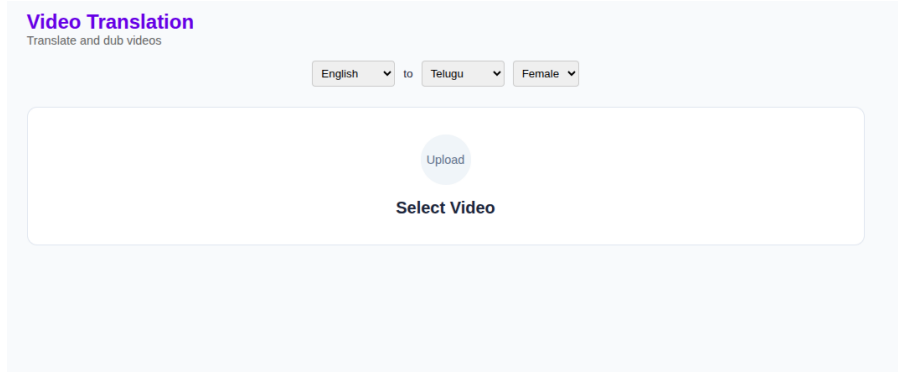


Figure 10: Video to Video translation

Users can inspect intermediate module outputs such as extracted source audio, timestamped transcription, translated text, and final translated videos. The website also supports synchronized comparison between the original and translated outputs to help evaluate dubbing quality, synchronization accuracy, and overall translation realism.

The website frontend and backend are currently developed and running successfully in the local environment. The system is designed using a modular architecture, allowing different models and pipeline components to be integrated easily at any stage of development. The codebase is structured to support flexible model replacement and future modifications without affecting the complete pipeline.

We are currently integrating additional modules into the Video-to-Video Translation pipeline, including synchronization and advanced processing components. Once all major modules are integrated and tested successfully, the complete system will be deployed and hosted for public access and further evaluation.

7.3 Future Enhancements

We are currently working on extending the pipeline with advanced features to improve synchronization, realism, and multilingual communication quality. One of the major future integrations is Wav2Lip-based lip synchronization, which will align translated speech with the speaker’s mouth movements to create realistic multilingual videos.

Another important challenge involves timestamp synchronization because translated speech may become shorter or longer depending on the target language. Different sentence structures and speaking speeds cause the translated audio to either compress or expand compared to the original video duration. To address this issue, we are exploring improved timestamp alignment techniques for maintaining synchronization between speech and video frames.

We are also studying emotion detection and tone mapping techniques to preserve emotional context during translation. By analyzing emotional speech features such as pitch, rhythm, and speaking style, future versions of the system can generate more expressive and realistic translated speech while maintaining the original speaker’s emotional intent.

8 Additional Challenges in Video-to-Video Translation

8.1 Face Detection Challenges

Face detection becomes difficult when videos contain low lighting conditions, blurred frames, or low-resolution inputs. In videos containing multiple people, the system may incorrectly detect or track the wrong face, leading to translation generation for the incorrect speaker. Continuous face tracking also becomes challenging when speakers move frequently or when camera angles change dynamically during the video.

8.2 Speaker Identification Challenges

In multi-speaker conversations, separating individual speaker voices becomes highly challenging, especially when speakers talk simultaneously or when background voices are present. Overlapping speech and environmental noise can confuse the system and result in incorrect voice extraction. This may further cause translated speech and lip synchronization to be assigned to the wrong speaker.

8.3 Code-Mixed Language Challenges

Real-world conversations often contain code-mixed speech where users combine multiple languages such as Hindi-English or Telugu-English within the same sentence. Automatic Speech Recognition systems may fail to identify language boundaries correctly, resulting in incorrect transcription and translation errors. This also creates synchronization mismatches in the final translated video output.

8.4 Text-to-Speech Challenges

Generating realistic translated speech is difficult because the system must preserve emotion, pitch, speaking speed, accent, and voice style of the original speaker. In many cases, synthesized speech may sound robotic or inconsistent, reducing the naturalness and realism of the translated video.

8.5 Voice Conversion Challenges

In multilingual translation systems, the translated speech should still preserve the original speaker’s voice identity. Maintaining the same voice characteristics across different languages is difficult because pronunciation patterns, phonemes, and speaking styles vary significantly between languages.

8.6 Error Propagation in the Pipeline

Video-to-Video Translation systems operate as connected multi-stage pipelines. Errors generated during earlier stages directly affect later stages. For example, incorrect transcription from Automatic Speech Recognition produces incorrect translation, which further affects Text-to-Speech generation and lip synchronization. These accumulated errors reduce the overall quality, synchronization, and realism of the final translated video.

9 Conclusion

In this project, we analyzed the complete Video-to-Video Translation pipeline and studied important modules such as Automatic Speech Recognition, Machine Translation, and Text-to-Speech systems. We explored major challenges including speaker diarization, gesture understanding, lip synchronization, emotion preservation, and multi-modal fusion through the analysis of various research papers and previous thesis work.

This study helped us understand how multilingual video translation systems maintain synchronization, speaker consistency, and communication realism across different stages of the pipeline. We also analyzed advanced approaches such as Wav2Lip, multi-modal learning, emotion-aware translation, and gesture-based feature extraction for future integration into our system.

The knowledge gained from this research provides a strong foundation for developing realistic, scalable, and efficient Video-to-Video Translation systems capable of supporting natural multilingual communication.

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